🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.2

Determine the Volume of Rectangular Prisms

Lesson 5-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

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CLASS LEGACY MISSION

Seal the Time Capsule

You are the lead builder for the school's 50-year time capsule project. Every memory must fit inside rectangular containers, and the principal needs to know exactly how much each one holds before they are buried. Master volume and the capsule gets sealed for the future.

My Notes
Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the volume of a rectangular prism with whole-number edges using length × width × height.

Language Objective: I can explain my work using the words volume, rectangular prism, cubic units, and edge.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Volume | Rectangular prism | Cubic units | Length, width, height | Net | Edge

👆 *Tap any word to see its meaning and a picture.*

1. The amount of space inside a three-dimensional solid is its ______________.

2. A solid with six rectangular faces, like a box, is a ______________.

3. The unit used to measure volume, like cubic centimeters, is ______________.

4. The three edge measurements of a rectangular prism are its ______________.

5. A flat pattern that folds up into a solid is a ______________.

6. A line segment where two faces of a solid meet is an ____________________.

Write About the Math

How many more cubic inches does the large box hold than the medium box?

¿Cuántas pulgadas cúbicas más tiene la caja grande que la caja mediana?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Volume** — *Volumen*
- ☐ **Rectangular prism** — *Prisma rectangular*
- ☐ **Cubic units** — *Unidades cúbicas*
- ☐ **Length, width, height** — *Largo, ancho, altura*
- ☐ **Net** — *Plantilla (desarrollo plano)*

Model: *Volume measures space in three directions, so you need length, width, and height.* — Students explain that volume is three-dimensional, so all three edges (length, width, height) are needed; two measurements only give area of one face.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The large box volume is ____ in³ because V = ____ × ____ × ____. The medium box volume is ____ in³. The large box holds ____ more cubic inches.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **Watch for this mistake:** *A common mistake in Volume with Whole Number Edges is multiplying only two of the three edges and stopping there — for example, seeing a box that is 6 in long, 4 in wide, and 3 in tall, multiplying just 6 × 4 = 24, and calling that the volume, when 24 is only the area of the bottom face (in square inches), not the volume of the whole box. The height must also be multiplied in: V = 6 × 4 × 3 = 72 cubic inches. A second common mistake is adding the three edges instead of multiplying them (6 + 4 + 3 = 13 instead of 6 × 4 × 3 = 72 cubic inches). Volume always MULTIPLIES all three edges, and the answer is always labeled in cubic units (in^3), never square units (in^2) or plain units (in).*
8. **Try It 1:** D. 30 cubic units — *V = length × width × height = 5 × 3 × 2 = 30 cubic units.*
9. **Try It 2:** D. 24 cubic units — *Each layer has 6 cubes and there are 4 layers, so 6 × 4 = 24 cubic units. This is the same as length × width × height with one factor being the number of layers.*